



Time Series Analysis

Seminar 9: Machine learning for time series



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Acronyms used in this material

- AI: Artificial Intelligence
- AR: AutoRegressive (model)
- ARIMA: AutoRegressive Integrated Moving Average (model)
- AUC: Area Under the (ROC) Curve
- BET: Bucharest Exchange Trading (index)
- CV: Cross-Validation
- DHR: Dynamic Harmonic Regression
- DM: Diebold–Mariano (test)
- ENTSO-E: European Network of Transmission System Operators for Electricity
- EODHD: EOD Historical Data (market-data provider)
- EU: European Union
- GB: Gradient Boosting (ensemble of regression trees)
- GW: Gigawatt
- HICP: Harmonised Index of Consumer Prices
- HLN: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- LSTM: Long Short-Term Memory (network)
- MAE: Mean Absolute Error
- MASE: Mean Absolute Scaled Error
- MIMO: Multiple-Input Multiple-Output (forecasting strategy)
- ML: Machine Learning
- MW: Megawatt
- OLS: Ordinary Least Squares
- RF: Random Forest
- RMSE: Root Mean Squared Error
- RW: Random Walk (forecast)
- SSE: Sum of Squared Errors

Today's question and route

- **Question:** how do we turn a time series into a machine-learning problem, and how do we know whether the result beats a simple benchmark?
 - ▶ this seminar comes **before** Lecture 9: the section “Prerequisites for Today” gives every definition the tasks use
- **Route**
 - ▶ Part A: a lag table and multi-step forecasts; leakage; one split of a tree; ridge and lasso in one dimension; a conformal interval; MASE and a Diebold–Mariano statistic
 - ▶ Part B: gradient boosting for the Romanian electricity load; recursive, direct and the value of holidays; Romanian inflation three months ahead; the sign of BET returns
 - ▶ Part C: a project idea and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	a lag table, recursive and direct forecasts; leakage in a rolling mean	Solved, Proposed	A1
A3, A4	one split of a regression tree; ridge and lasso in one dimension	Solved, Proposed	A3
A5, A6	a conformal interval and the pinball loss; MASE and a DM statistic	Solved, Proposed	A5
B1, B2	electricity load: boosting against the seasonal naive; recursive, direct, holidays	Solved, Proposed	B1
B3, B4	Romanian inflation; the sign of BET returns	Solved, Proposed	B3
C1, C2	a global model as a project; what is wrong in an AI answer?	Proposed	B1, B3

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
Electricity load, Romania	ENTSO-E	hourly, averaged per day (GW)	2022–2026
HICP, 27 EU countries	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	2005–2026
BET	EODHD	daily, close	2000–2026

- ENTSO-E: the European Network of Transmission System Operators for Electricity; the same load file as in [Chapter 4](#)
- In the notebook: `direct_frame`, `forecast_direct`, `forecast_recursive`, `dm_raw`, `inflation_forecasts`, `sign_frame`; scikit-learn models with fixed seeds; no account or key is needed

Prerequisites for Today (1/5): from a series to a table

- **Supervised learning:** learn $\hat{f}$ from pairs (features x_i , target y_i) so that $\hat{f}(x)$ is close to y on **new** data
 - ▶ for a series: at the origin t , features = lags $y_t, y_{t-1}, \dots$, rolling means of past values, calendar of the target day; target y_{t+h} , h : the horizon
- **Multi-step strategies**
 - ▶ recursive: one model for $h = 1$, applied again with its own forecasts as inputs
 - ▶ direct: one model for each horizon h , trained on the target y_{t+h}
 - ▶ MIMO (multiple-input multiple-output): one model with H outputs, H : the longest horizon
- A series of length n with p lags gives $n - p - h + 1$ complete rows for horizon h

Prerequisites for Today (2/5): validation and leakage

- ▣ **Walk-forward** validation: train on data up to the origin, forecast the next block, move the origin; errors pooled over all origins
 - ▶ never random K -fold for forecasting models with overlapping targets or exogenous features
- ▣ **Leakage**: information unavailable at the origin enters the features or the training
 - ▶ a rolling mean that contains y_{t+1} ; scaling fitted on the full sample; overlapping targets without a gap; actual instead of forecast weather
 - ▶ calendar variables (weekday, holidays) are known in advance: no leak
- ▣ Benchmarks ([Chapter 0](#)): naive $\hat{y}_{T+h} = y_T$, seasonal naive $\hat{y}_{T+h} = y_{T+h-m}$
 - ▶ T : the forecast origin (last observation); m : the seasonal period ($m = 7$ for daily data: the same weekday last week)

Prerequisites for Today (3/5): ridge and lasso

- ▣ Penalised least squares: fit plus a penalty that shrinks the coefficients β_j towards 0
 - ▶ **ridge**: $\min \sum_t (y_t - \mathbf{x}_t' \beta)^2 + \lambda \sum_j \beta_j^2$; **lasso**: penalty $\lambda \sum_j |\beta_j|$ (some $\hat{\beta}_j = 0$)
 - ▶ $\mathbf{x}_t$: the features of row t ; $\lambda \geq 0$: the strength of the penalty ($\lambda = 0$: OLS)
- ▣ One standardised feature (mean 0, standard deviation 1): closed forms
 - ▶ $S_{xy} = \sum \mathbf{x}_t y_t$, $S_{xx} = \sum \mathbf{x}_t^2$; OLS S_{xy}/S_{xx} ; ridge $S_{xy}/(S_{xx} + \lambda)$
 - ▶ lasso $\text{sign}(S_{xy}) \max(|S_{xy}| - \lambda/2, 0)/S_{xx}$: exactly 0 when $\lambda/2 \geq |S_{xy}|$ (soft thresholding)

Prerequisites for Today (4/5): trees and ensembles

- ▣ **Regression tree:** split " $x \leq c$ "; each side predicts the mean of its targets
 - ▶ c : the threshold; the best c minimises the total SSE (sum of squared errors around the two means)
 - ▶ candidates: midpoints between consecutive sorted values of x ; a tree cannot forecast outside the range of the training targets
- ▣ **Random forest:** average of many deep trees on bootstrap samples
 - ▶ **gradient boosting** (GB): small trees added one by one on the residuals, each scaled by a learning rate $\nu \in (0, 1]$
 - ▶ scikit-learn: `HistGradientBoostingRegressor`; the same family as LightGBM and XGBoost

Prerequisites for Today (5/5): evaluation and intervals

- **MASE** = MAE / (in-sample MAE of the seasonal naive method); MAE: mean absolute error; below 1: better than that benchmark
 - ▶ **Diebold–Mariano** ([Chapter 4](#)): $d_t = L(e_{1t}) - L(e_{2t})$, $DM = \bar{d} / \sqrt{\hat{s}^2/n}$; $|DM|$ above the critical value: different accuracy
 - ▶ e_{1t} , e_{2t} : the errors of the two models; L : the loss (absolute or squared); $\bar{d}$, $\hat{s}^2$: mean and variance of d_t ; n : number of origins
- **Pinball loss** of a τ -quantile forecast q : $\tau(y - q)$ if $y \geq q$, else $(1 - \tau)(q - y)$
 - ▶ **split conformal** interval: $\hat{y} \pm \hat{q}$, $\hat{q} =$ the $\lceil (n + 1)(1 - \alpha) \rceil$ -th smallest of n absolute calibration errors; α : allowed miscoverage; $\lceil \cdot \rceil$: rounding up
- **Classification** of the sign: the baseline is the majority class of the training data, not 50%
 - ▶ $z = (\text{acc} - \text{acc}_0) / \sqrt{\text{acc}_0(1 - \text{acc}_0)/n}$ compares the accuracy acc with the baseline accuracy acc_0 on n test days

A1: a lag table and two strategies [Solved]

Task

- ▣ Series $y_1, \dots, y_8$: 10, 12, 11, 13, 14, 13, 15, 16.
- ▣ 1. Build the table with the features y_{t-1} , y_{t-2} and the target y_t ; how many rows does it have?
- ▣ 2. With the fitted model $\hat{y}_t = 2 + 0.6 y_{t-1} + 0.3 y_{t-2}$, compute the recursive forecasts $\hat{y}_9$, $\hat{y}_{10}$, $\hat{y}_{11}$.
- ▣ 3. A direct two-step model is $\hat{y}_{t+2} = 3 + 0.5 y_t + 0.3 y_{t-1}$: compute $\hat{y}_{10}$ and the number of rows of its table.
- ▣ Report: the table, three recursive forecasts, one direct forecast, two row counts.

Solution

t	y_{t-1}	y_{t-2}	y_t
3	12	10	11
4	11	12	13
5	13	11	14
6	14	13	13
7	13	14	15
8	15	13	16

- ▣ 1. 6 rows ($t = 3, \dots, 8$)
- ▣ 2. $\hat{y}_9 = 2 + 0.6 \cdot 16 + 0.3 \cdot 15 = 16.10$;
 $\hat{y}_{10} = 2 + 0.6 \cdot 16.10 + 0.3 \cdot 16 = 16.46$; $\hat{y}_{11} = 16.71$
- ▣ 3. $\hat{y}_{10} = 3 + 0.5 \cdot 16 + 0.3 \cdot 15 = 15.50$; 5 rows (origins $t = 2, \dots, 6$)

A2: spot the leak [Proposed]

Solution discussed in class

Task

- ▣ Daily series 10, 12, 11, 13, 14; the target is the last value, 14. Model: A1.
- ▣ 1. Compute the correct 3-day mean feature for this target, and the mean that a careless `rolling(3).mean()` without `shift(1)` would give.
- ▣ 2. For each choice, say whether it leaks: (a) a scaler fitted on 2022–2026 before the split; (b) holiday dummies of the target day; (c) the temperature measured on the target day; (d) 5-fold shuffled CV for a 21-day target.
- ▣ Report: two means and four verdicts with one reason each.

A3: one split of a regression tree [Solved]

Task

- Six days: x = load on the same weekday last week, y = load today (GW): (4.8, 5.0), (5.0, 5.2), (5.6, 5.9), (6.0, 6.0), (6.4, 6.6), (6.9, 7.0).
- 1. List the five candidate thresholds.
- 2. For each, compute the two leaf means and the total SSE.
- 3. Choose the split and give the forecast for $x = 6.2$.
- Report: a table of five rows, the chosen threshold, one forecast.

Solution

threshold	left mean	right mean	SSE
4.9	5.00	6.14	1.912
5.3	5.10	6.38	0.827
5.8	5.37	6.53	0.953
6.2	5.53	6.80	0.828
6.7	5.74	7.00	1.672

- Without a split: mean 5.95, SSE 2.995
- Best: $x \leq 5.3$, SSE 0.827; leaves 5.10 and 6.38 GW
- $x = 6.2 > 5.3$: forecast 6.38 GW

A4: ridge and lasso in one dimension [Proposed]

Solution discussed in class

Task

- ▣ One standardised feature with $S_{xy} = 40$ and $S_{xx} = 50$. Model: A3 and the formulas of "Prerequisites for Today (3/5)".
- ▣ 1. Compute the OLS coefficient.
- ▣ 2. Compute the ridge coefficient for $\lambda = 10$ and $\lambda = 50$.
- ▣ 3. Compute the lasso coefficient for $\lambda = 20$ and $\lambda = 100$.
- ▣ 4. Explain in one sentence which method can drop a lag from the model.
- ▣ Report: five coefficients and one sentence.

A5: a conformal interval and the pinball loss [Solved]

Task

- ▣ A model forecasts tomorrow's load $\hat{y} = 6.10$ GW. On 9 calibration days its absolute errors were 0.12, 0.30, 0.05, 0.22, 0.41, 0.18, 0.09, 0.27, 0.35 GW.
- ▣ 1. Build the 80% split conformal interval.
- ▣ 2. A 90% quantile forecast is $q = 6.4$ GW: compute its pinball loss if $y = 6.0$ and if $y = 6.6$.
- ▣ Report: k , $\hat{q}$, the interval, two losses.

Solution

- ▣ 1. Sorted: 0.05, 0.09, 0.12, 0.18, 0.22, 0.27, 0.30, 0.35, 0.41;
 $k = \lceil 10 \cdot 0.8 \rceil = 8$; $\hat{q} = 0.35$
- ▣ interval $6.10 \pm 0.35 = [5.75, 6.45]$ GW
- ▣ 2. $y = 6.0 < q$: $(1 - 0.9)(6.4 - 6.0) = 0.04$; $y = 6.6 > q$:
 $0.9 \cdot 0.2 = 0.18$
- ▣ an outcome above a high quantile costs nine times more per GW: the loss pushes q up until only 10% of outcomes exceed it

A6: MASE and a Diebold–Mariano statistic [Proposed]

Solution discussed in class

Task

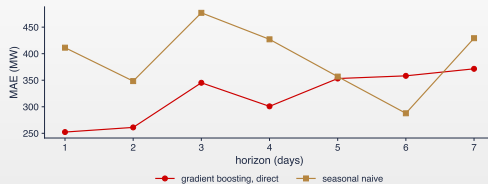
- ▣ Average absolute errors (GW) of two models at 6 origins: $A = 0.30, 0.25, 0.40, 0.28, 0.35, 0.22$; $B = 0.32, 0.30, 0.38, 0.35, 0.41, 0.30$; MASE scale 0.31 GW. Model: A5 and “Prerequisites for Today (5/5)”.
- ▣ 1. Compute the MAE and the MASE of each model.
- ▣ 2. Compute $d_t = A_t - B_t$, its mean and standard deviation, and $DM = \bar{d}/(s/\sqrt{6})$.
- ▣ 3. Compare $|DM|$ with the 5% critical value of $t(5)$ and conclude.
- ▣ Report: four accuracy numbers, $\bar{d}$, DM and a decision.



B1: gradient boosting for the electricity load [Solved]

- **Question:** does a gradient-boosting model forecast next week's daily load better than "the same weekday last week"?
- **Inputs:** daily mean load of Romania (GW), 2022–2026; weekly origins from 5 January to 22 June 2026; horizons 1–7 days
- **Tasks**
 1. Build the direct tables (14 lags, 7- and 28-day means, the same weekday, calendar of the target day) with `direct_frame`.
 2. At each origin, train one `HistGradientBoostingRegressor` per horizon on the data before the origin and forecast the next 7 days.
 3. Compute the MAE, RMSE and MASE of GB and of the seasonal naive forecast.
 4. Run the Diebold–Mariano test on the origin-average absolute errors.
 5. Interpretation: at which horizons does GB gain most, and why?
- **Report:** six accuracy numbers, a DM statistic with its p-value, one sentence
- **Notebook:** section B1

B1: solution [Solved]



- 25 origins, 7 horizons; MAE: GB 320 MW, seasonal naive 391 MW; RMSE 409 and 521 MW; MASE 1.04 and 1.27 (scale 308 MW)
- DM (with the HLN correction) = -2.63 , p 0.015; the worst origin for GB: 5 January 2026 (MAE 576 MW)
- Interpretation: one day ahead GB has 252 MW against 411 MW: it uses yesterday's level, which the seasonal naive ignores; at 7 days (371 and 429 MW) both rely on last week's pattern

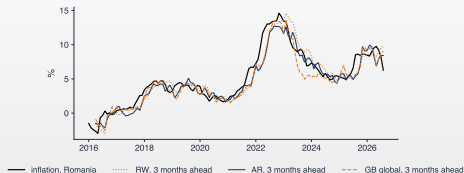
B2: recursive, direct and the value of holidays [Proposed]

- ▣ **Question:** does the recursive strategy forecast as well as the direct one, and how much do the holiday features help?
- ▣ **Inputs:** the data and origins of B1; model: B1
- ▣ **Tasks**
 1. Repeat B1 with the recursive strategy (`forecast_recursive`).
 2. Repeat the direct model without the three holiday features (`easter`, `xmas`, `holiday`).
 3. Report the MAE of the three models on all days, on the public-holiday days and on the other days.
 4. Interpretation: which feature group explains most of the difference, and on which days?
- ▣ **Report:** nine MAE values and one sentence
- ▣ **Notebook:** section B2

B3: Romanian inflation three months ahead [Solved]

- **Question:** can machine learning forecast Romanian inflation three months ahead better than the random walk and a linear AR model?
- **Inputs:** 12-month HICP inflation of 27 EU countries (Eurostat), monthly, 2005–2026
- **Tasks**
 1. Build the features of each country (inflation lags, monthly changes, calendar month) and the target $\pi_{t+3} - \pi_t$ with `inflation_frame`.
 2. Every January from 2016, refit AR, lasso, random forest and GB on Romania only, and a global GB on all countries; forecast every month of the year.
 3. Compute the RMSE of each model and its ratio to the random walk.
 4. Run the DM test for AR against RW and for the global against the local GB.
 5. Interpretation: what happened to all models in 2021–2023?
- **Report:** six RMSEs, two DM tests and one sentence
- **Notebook:** section B3

B3: solution [Solved]



- ▣ 125 origins, January 2016–May 2026; RMSE (pp): RW 1.41, AR 1.28 (0.90), lasso 1.45 (1.02), RF 1.49 (1.06), GB local 1.64 (1.16), GB global 1.37 (0.97)
- ▣ DM: AR against RW -1.33 (p 0.186); GB global against GB local -1.60 (p 0.113)
- ▣ Interpretation: from mid-2021 to mid-2023 every RMSE jumps (RW 2.27, AR 2.02, GB global 2.09 pp): the energy shock was not in the past of inflation; pooling 27 countries helps the trees, but no model is significantly better than AR

B4: the sign of BET returns [Proposed]

- ▣ **Question:** can a classifier predict whether the BET rises tomorrow better than “it always rises”?
- ▣ **Inputs:** daily BET log returns since 2000; test years 2015–2026; model: B3 (yearly refits)
- ▣ **Tasks**
 1. Build the features of `sign_frame` (five lagged returns, sums over 5, 21, 63 days, volatility over 21 and 63 days) and the target $1\{r_{t+1} > 0\}$.
 2. Each January, fit a logistic regression and a gradient-boosting classifier on all earlier days; the baseline predicts the majority class of the training data.
 3. Report the accuracy of the three, the z statistic against the baseline and the AUC.
 4. Interpretation: what does the result say about the weak-form efficiency of the Bucharest market?
- ▣ **Report:** three accuracies, two z statistics, two AUCs and one sentence
- ▣ **Notebook:** section B4

C1: a global model for European electricity load [Proposed]

- **Idea:** ENTSO-E publishes the hourly load of every European country; one global GB model trained on 20 countries may forecast Romania better than a local one
 - ▶ features: lags scaled by each country's mean, weekday, national holidays, a country identifier
- Questions to answer
 - ▶ 1. Does the global model beat the local GB and DHR on the 26 origins of [Chapter 4](#)?
 - ▶ 2. Does it forecast Orthodox Easter better, although most countries celebrate Western Easter?
 - ▶ 3. Which countries help Romania most (leave-one-country-out)?
- A possible team project: data, code, DM tests and a two-page report; model: B1, B3

C2: audit an AI answer [Proposed]

- A student asked an AI assistant whether machine learning can forecast markets and electricity load. The answer:
- (a) A random forest with 5-fold cross-validation explains 36.8% of the next month's S&P 500 return, so the market is predictable.
- (b) A logistic model predicts the daily sign with 53.8% accuracy, well above 50%: a profitable signal.
- (c) For the load, GB has MASE 0.97, so it is 3% more accurate than DHR.
- (d) Standardise the whole series first, then split it into training and test.
- (e) An LSTM always beats ARIMA because it has long memory.
- (f) Calendar features of the target day, such as holidays, are allowed because they are known in advance.
- Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the lecture notebook (sections 2, 6, 9).
 - ▶ 2. Report: a list of six verdicts with one line of justification each.

Key takeaways

- ▣ A forecast is a prediction from features known at the origin: lags, past rolling means, calendar
- ▣ Recursive and direct strategies: compare them on the same origins
- ▣ Leakage: any value from after the origin, including through scaling or shuffled folds
- ▣ GB beats the seasonal naive forecast for the load; nothing beats “always up” for the sign of BET returns
- ▣ An AI answer is a draft: check the validation scheme and the benchmark behind every claimed accuracy

After the seminar

- Lecture 9 develops each topic of today: features and strategies, walk-forward validation, ridge and lasso, trees and boosting, neural networks and the LSTM, prediction intervals, global models, the M4 and M5 competitions
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: a global model for European electricity load
- Reading: [Huang and Petukhina \(2022\)](#), Ch. 10; [Hyndman and Athanasopoulos \(2021\)](#), Sec. 12.4; [Montero-Manso and Hyndman \(2021\)](#)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

References

- ▣ Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. doi:10.1080/07350015.1995.10524599
- ▣ Huang, C., & Petukhina, A. (2022). *Applied Time Series Analysis and Forecasting with Python*. Springer. doi:10.1007/978-3-031-13584-2
- ▣ Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice* (3rd ed.), Section 12.4, Neural network models. OTexts. otexts.com/fpp3/nnetar.html
- ▣ Hyndman, R. J., & Koehler, A. B. (2006). Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4), 679–688. doi:10.1016/j.ijforecast.2006.03.001
- ▣ Kaufman, S., Rosset, S., Perlich, C., & Stitelman, O. (2012). Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4), 1–21. doi:10.1145/2382577.2382579
- ▣ Koenker, R., & Bassett, G. (1978). Regression quantiles. *Econometrica*, 46(1), 33–50. doi:10.2307/1913643
- ▣ Lei, J., G'Sell, M., Rinaldo, A., Tibshirani, R. J., & Wasserman, L. (2018). Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523), 1094–1111. doi:10.1080/01621459.2017.1307116
- ▣ Montero-Manso, P., & Hyndman, R. J. (2021). Principles and algorithms for forecasting groups of time series: Locality and globality. *International Journal of Forecasting*, 37(4), 1632–1653. doi:10.1016/j.ijforecast.2021.03.004